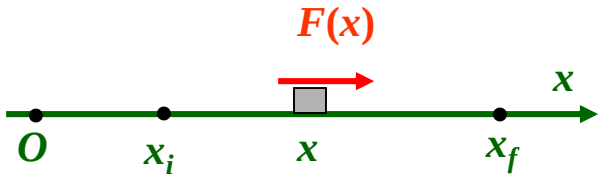
Chapter 24
Electric Potential

In this chapter we will define the **electric potential** **(symbol *V*)** associated with the electric force and accomplish the following tasks:

Calculate *V* if we know the corresponding electric field.
Calculate the electric field if we know the corresponding potential *V*.
Determine the potential *V* generated by a point charge.
Determine the potential *V* generated by a discrete charge distribution.
Determine the potential *V* generated by a continuous charge distribution.
Determine the electric potential energy *U* of a system of charges.
Define the notion of an equipotential surface.
Explore the geometric relationship between equipotential surfaces and electric field lines.
Explore the potential of a charged isolated conductor.

$$\Delta U = -\int_{x_i}^{x_f} F(x)dx$$



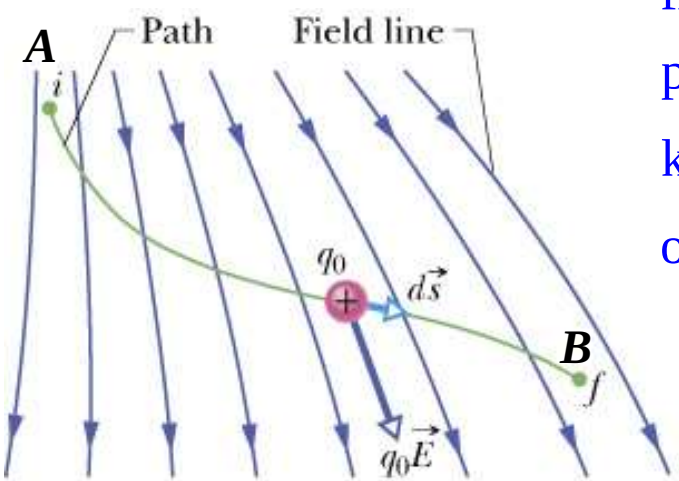
Electric Potential Energy

In Chapter 8 we defined the **change in potential energy** ΔU associated with a conservative force as the negative value of the work W that the force must do on a particle to take it from an initial position x_i to a final position x_f.

$$\Delta U = U_f - U_i = -W = -\int_{x_i}^{x_f} F(x)dx$$

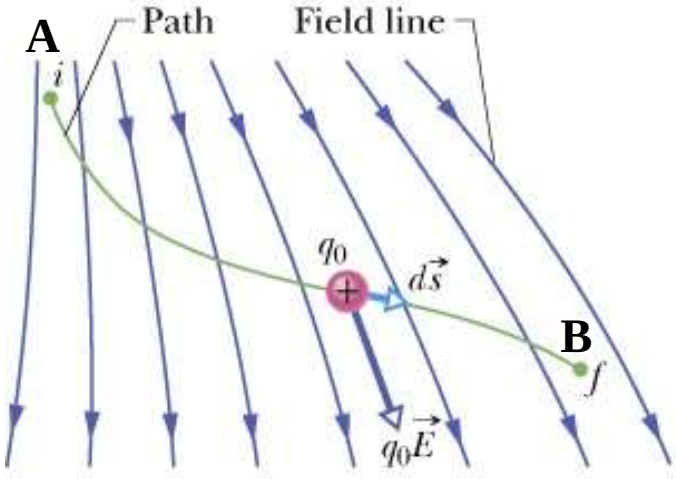
$$\Delta U = -q_0 \int_i^f \vec{E} \cdot d\vec{s}$$

Consider an electric charge q_0 moving from an initial position at point A to a final position at point B under the influence of a known electric field $\vec{E}$. The force exerted on the charge is $\vec{F} = q_0 \vec{E}$.



$$\Delta U = -\int_i^f \vec{F} \cdot d\vec{s} = -q_0 \int_i^f \vec{E} \cdot d\vec{s} \qquad \textbf{(24-2)}$$

$$V_P = -\int_{\infty}^{P} \vec{E} \cdot d\vec{s}$$



The Electric Potential V

The change in potential energy of a charge q_0 moving under the influence of $\vec{E}$ from point A to point B is: $\Delta U = U_f - U_i = -W = -q_0 \int_i^f \vec{E} \cdot d\vec{s}.$

Please note that ΔU depends on the value of q_0.

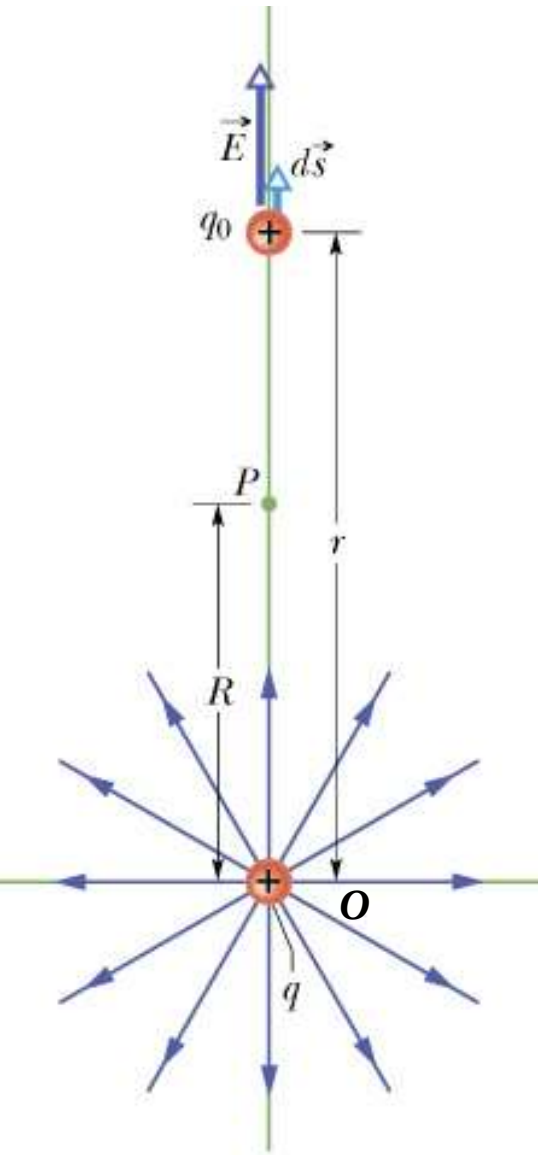
We define the **electric potential** V in such a manner so that it is independent of q_0: $\Delta V = \dfrac{\Delta U}{q_0} = -\dfrac{W}{q_0}$ Here $\Delta V = V_f - V_i \rightarrow V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}.$

In all physical problems only changes in V are involved. Thus we can define arbitrarily the value of V at a reference point, which we choose to be at infinity: $V_f = V_\infty = 0$. We take the initial position as the generic point P with potential V_P :

$V_P = -\int_{\infty}^{P} \vec{E} \cdot d\vec{s}.$ The potential V_P depends only on the coordinates of P and on $\vec{E}$.

(24-3)

$$V_P = \frac{1}{4\pi\varepsilon_0}\frac{q}{R}$$



SI Units of V : Definition of voltage : $\Delta V = -\dfrac{W}{q_0}$

Units of V: J/C, known as the volt

Potential Due to a Point Charge

Consider a point charge q placed at the origin. We will use the definition given on the previous page to determine the potential V_P at point P a distance R from O.

$$V_P = -\int_{\infty}^{R} \vec{E}\cdot d\vec{s} = \int_{R}^{\infty} E dr \cos 0 = \int_{R}^{\infty} E dr$$

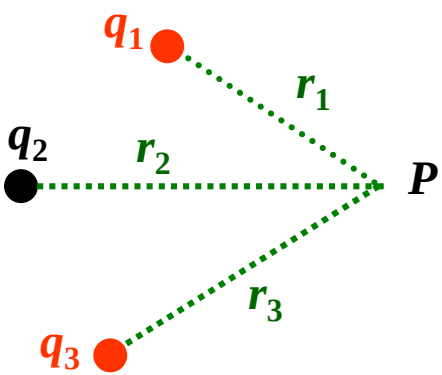
The electric field generated by q is:

$$E = \frac{q}{4\pi\varepsilon_0 r^2}$$

$$V_P = \frac{q}{4\pi\varepsilon_0}\int_{R}^{\infty}\frac{dr}{r^2} \qquad \int\frac{dr}{x^2} = -\frac{1}{x}$$

$$\rightarrow V_P = \frac{q}{4\pi\varepsilon_0}\left[-\frac{1}{r}\right]_R^{\infty} = \frac{1}{4\pi\varepsilon_0}\frac{q}{R}$$

(24-4)



Potential Due to a Group of Point Charges

Consider the group of three point charges shown in the figure. The potential V generated by this group at any point P is calculated using the principle of superposition.

1. We determine the potentials $V_1, V_2,$ and V_3 generated by each charge at point P:

$$\boxed{V = V_1 + V_2 + V_3}$$

$$V_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1}, \quad V_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2}, \quad V_3 = \frac{1}{4\pi\varepsilon_0}\frac{q_3}{r_3}$$

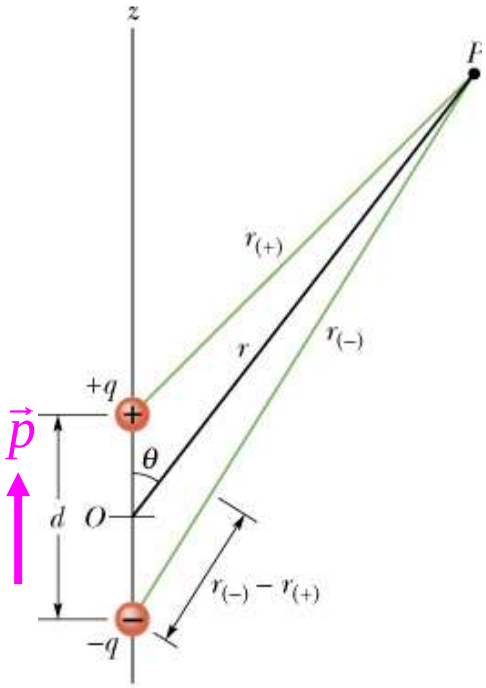
2. We add the three terms:

$$V = V_1 + V_2 + V_3$$

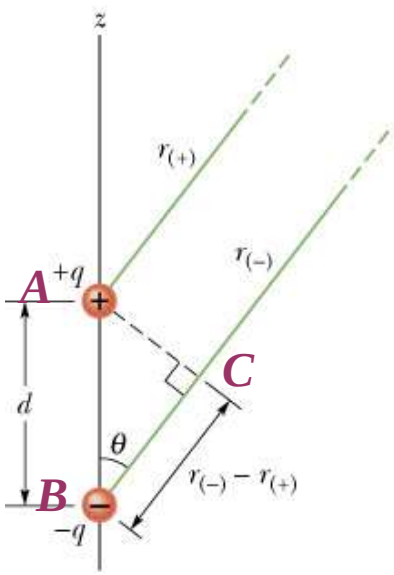
$$V = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1} + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2} + \frac{1}{4\pi\varepsilon_0}\frac{q_3}{r_3}$$

The previous equation can be generalized for n charges as follows:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1} + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2} + \ldots + \frac{1}{4\pi\varepsilon_0}\frac{q_n}{r_n} = \frac{1}{4\pi\varepsilon_0}\sum_{1}^{n}\frac{q_i}{r_i} \qquad \textbf{(24-5)}$$



(a)



(b)

Example : Potential Due to an Electric Dipole

Consider the electric dipole shown in the figure.

We will determine the electric potential V created at point P by the two charges of the dipole using superposition.

Point P is at a distance r from the center O of the dipole.

Line OP makes an angle θ with the dipole axis:

$$V = V_{(+)} + V_{(-)} = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_{(+)}} - \frac{q}{r_{(-)}}\right) = \frac{q}{4\pi\varepsilon_0}\frac{r_{(-)} - r_{(+)}}{r_{(-)}r_{(+)}}.$$

We assume that $r \gg d$, where d is the charge separation.

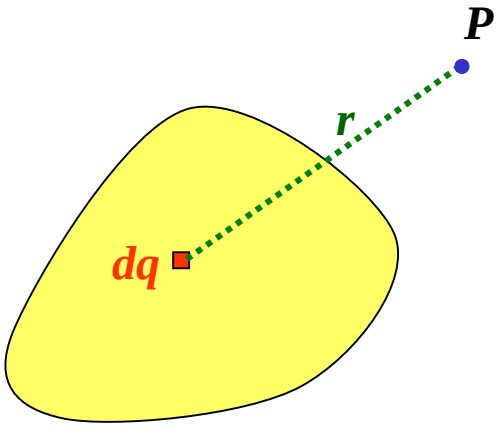
From triangle ABC we have: $r_{(-)} - r_{(+)} \approx d\cos\theta$.

Also: $r_{(-)}r_{(+)} \approx r^2 \rightarrow V \approx \dfrac{q}{4\pi\varepsilon_0}\dfrac{d\cos\theta}{r^2} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{p\cos\theta}{r^2}$,

where $p = qd =$ the electric dipole moment.

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2} \qquad \textbf{(24-6)}$$

Potential Due to a Continuous Charge Distribution



Consider the charge distribution shown in the figure. In order to determine the electric potential V created by the distribution at point P we use the principle of superposition as follows:

1. We divide the distribution into elements of charge dq.

For a volume charge distribution, $dq = \rho dV$.

For a surface charge distribution, $dq = \sigma dA$.

For a linear charge distribution, $dq = \lambda d\ell$.

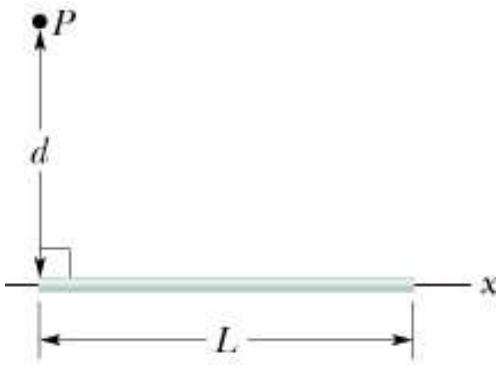
$$\boxed{V = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r}}$$

2. We determine the potential dV created by dq at P: $\quad dV = \dfrac{1}{4\pi\varepsilon_0}\dfrac{dq}{r}$.

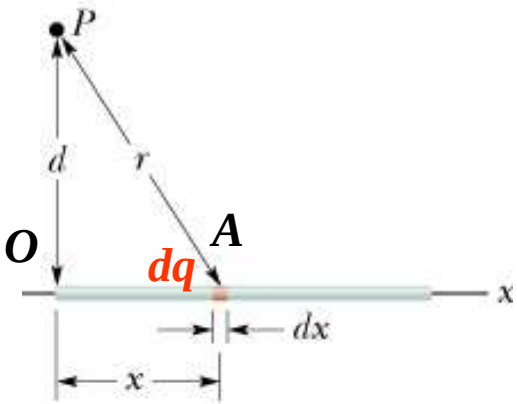
3. We sum all the contributions in the form of the integral: $\quad V = \dfrac{1}{4\pi\varepsilon_0}\displaystyle\int \dfrac{dq}{r}$.

Note 1 : The integral is taken over the whole charge distribution.

Note 2 : The integral involves only scalar quantities.



(a)



(b)

Example : Potential created by a line of charge of length L and uniform linear charge density λ at point P. Consider the charge element $dq = \lambda dx$ at point A, a distance x from O. From triangle OAP we have:

$r = \sqrt{d^2 + x^2}$. Here d is the distance OP.

The potential dV created by dq at P is:

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dx}{\sqrt{d^2 + x^2}}$$

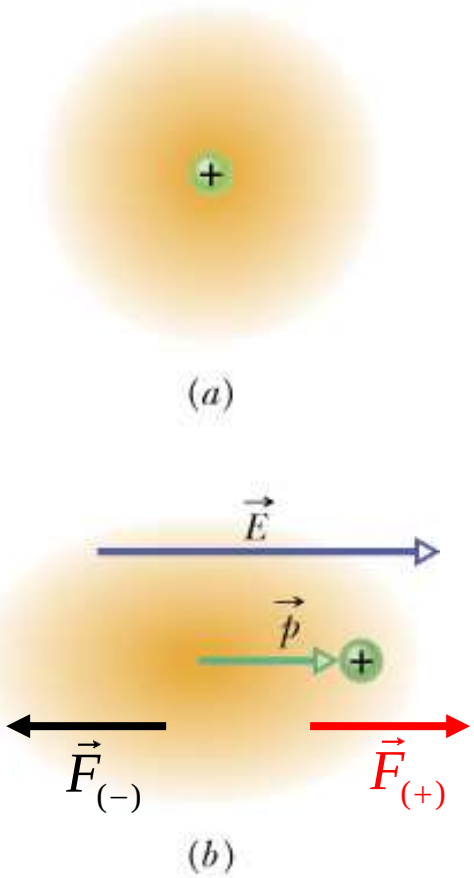
$$V = \frac{\lambda}{4\pi\varepsilon_0}\int_0^L \frac{dx}{\sqrt{d^2 + x^2}}$$

$$\int \frac{dx}{\sqrt{d^2 + x^2}} = \ln\left(x + \sqrt{d^2 + x^2}\right)$$

$$V = \frac{\lambda}{4\pi\varepsilon_0}\left[\ln\left(x + \sqrt{d^2 + x^2}\right)\right]_0^L$$

$$V = \frac{\lambda}{4\pi\varepsilon_0}\left[\ln\left(L + \sqrt{L^2 + x^2}\right) - \ln d\right] \qquad \textbf{(24-8)}$$

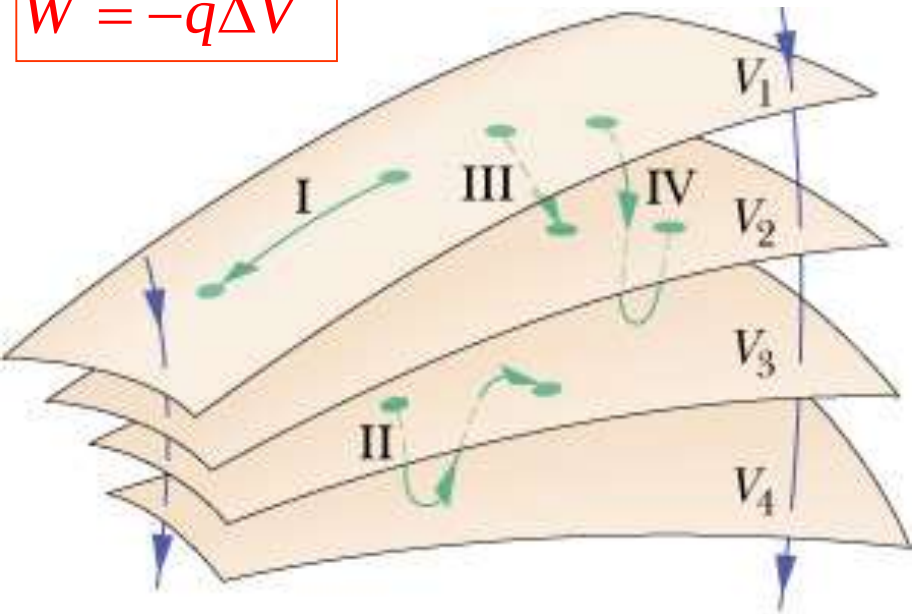
Induced Dipole Moment



Many molecules such as H_2O have a permanent electric dipole moment. These are known as "polar" molecules. For others, such as O_2 , N_2, etc. the electric dipole moment is zero. These are known as "nonpolar" molecules. One such molecule is shown in fig. *a*. The electric dipole moment $\vec{p}$ is zero because the center of the positive charge coincides with the center of the negative charge. In fig. *b* we show what happens when an electric field $\vec{E}$ is applied to a nonpolar molecule. The electric forces on the positive and negative charges are equal in magnitude but opposite in direction.

As a result the centers of the positive and negative charges move in opposite directions and do not coincide. Thus a nonzero electric dipole moment $\vec{p}$ appears. This is known as "induced" electric dipole moment, and the molecule is said to be "polarized." When the electric field is removed $\vec{p}$ disappears.

$$W = -q\Delta V$$



Equipotential Surfaces

A collection of points that have the same potential is known as an equipotential surface. Four such surfaces are shown in the figure. The work done by $\vec{E}$ as it moves a charge q between two points that have a potential difference ΔV is given by $W = -q\Delta V$.

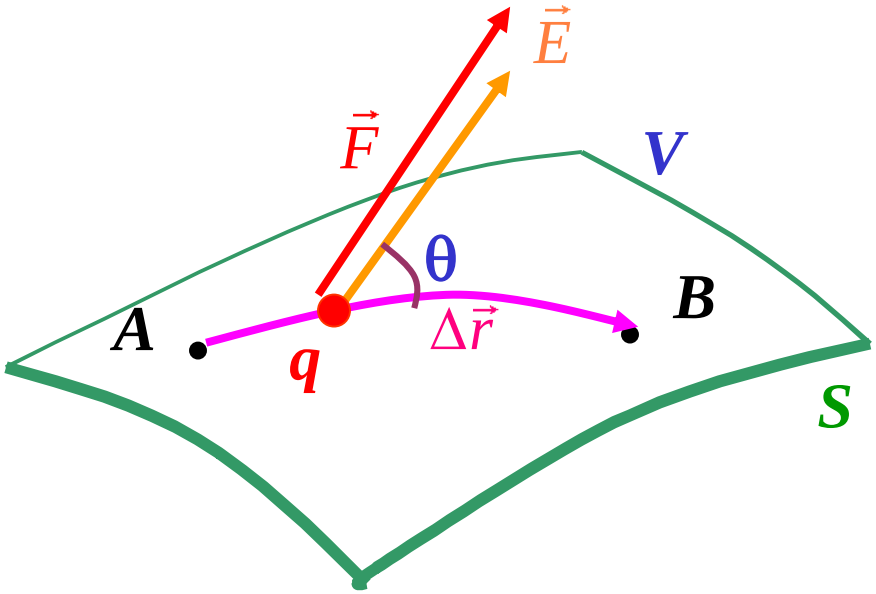
For path I : $W_{\mathrm{I}} = 0$ because $\Delta V = 0$.

For path II: $W_{\mathrm{II}} = 0$ because $\Delta V = 0$.

For path III: $W_{\mathrm{III}} = q\Delta V = q\left(V_2 - V_1\right)$.

For path IV: $W_{\mathrm{IV}} = q\Delta V = q\left(V_2 - V_1\right)$.

Note : When a charge is moved on an equipotential surface $\left(\Delta V = 0\right)$ the work done by the electric field is zero: $W = 0$.



The Electric Field $\vec{E}$ is Perpendicular to the Equipotential Surfaces

Consider the equipotential surface at potential V. A charge q is moved by an electric field $\vec{E}$ from point A to point B along a path $\Delta\vec{r}$. Points A and B and the path lie on S.

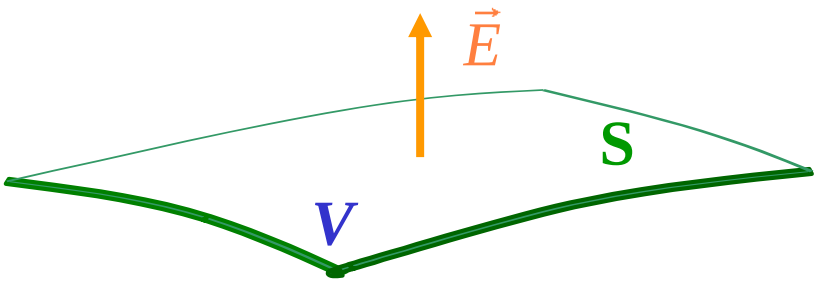
Let's assume that the electric field $\vec{E}$ forms an angle θ with the path $\Delta\vec{r}$.

The work done by the electric field is: $W = \vec{F}\cdot\Delta\vec{r} = F\Delta r\cos\theta = qE\Delta r\cos\theta.$

We also know that $W = 0$. Thus: $qE\Delta r\cos\theta = 0,$ where

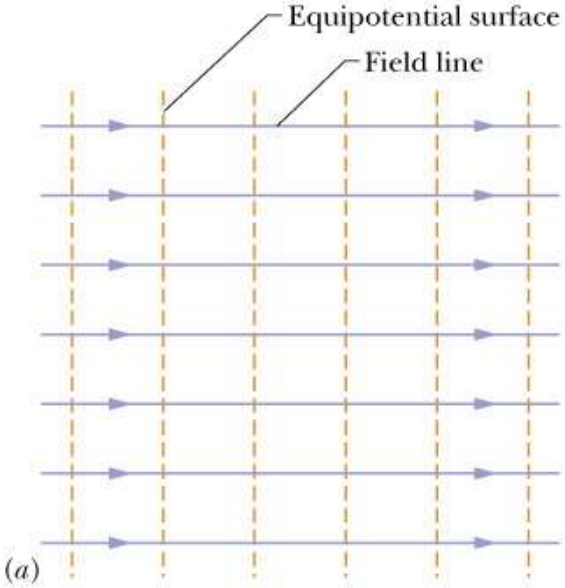
$q \neq 0, \quad E \neq 0, \quad \Delta r \neq 0.$ Thus: $\cos\theta = 0 \quad \rightarrow \quad \theta = 90°.$

The correct picture is shown in the figure below.

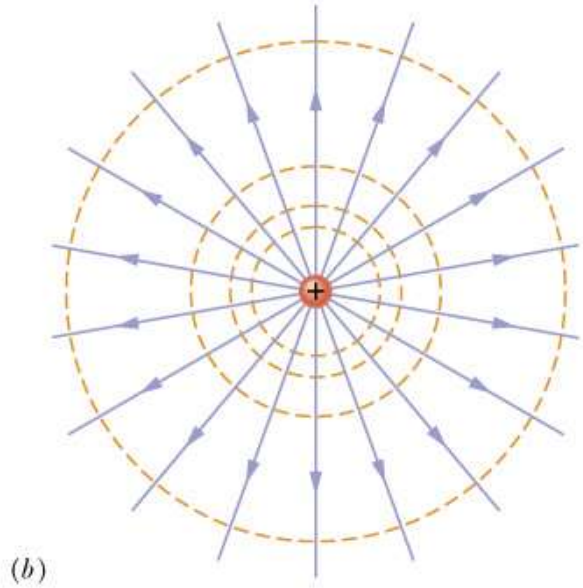


Examples of Equipotential Surfaces and the Corresponding Electric Field Lines

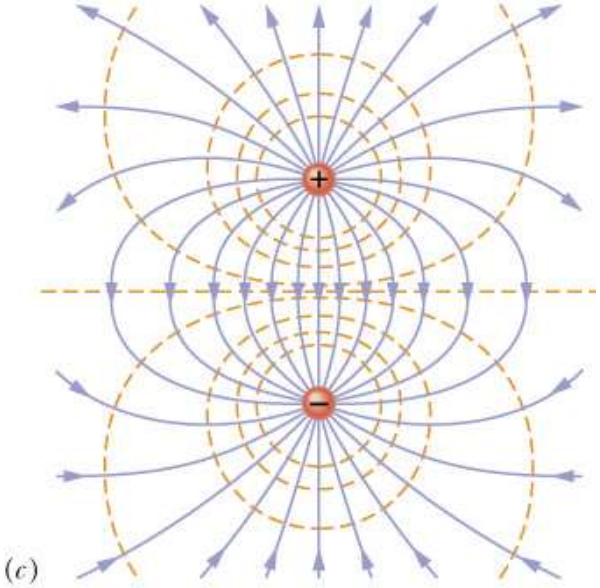
Uniform electric field



Isolated point charge



Electric dipole



Equipotential surfaces for a point charge *q* :

$V = \dfrac{q}{4\pi\varepsilon_0 r}$. Assume that V is constant $\rightarrow r = \dfrac{q}{4\pi\varepsilon_0 V} =$ constant.

Thus the equipotential surfaces are spheres with their center at the point charge

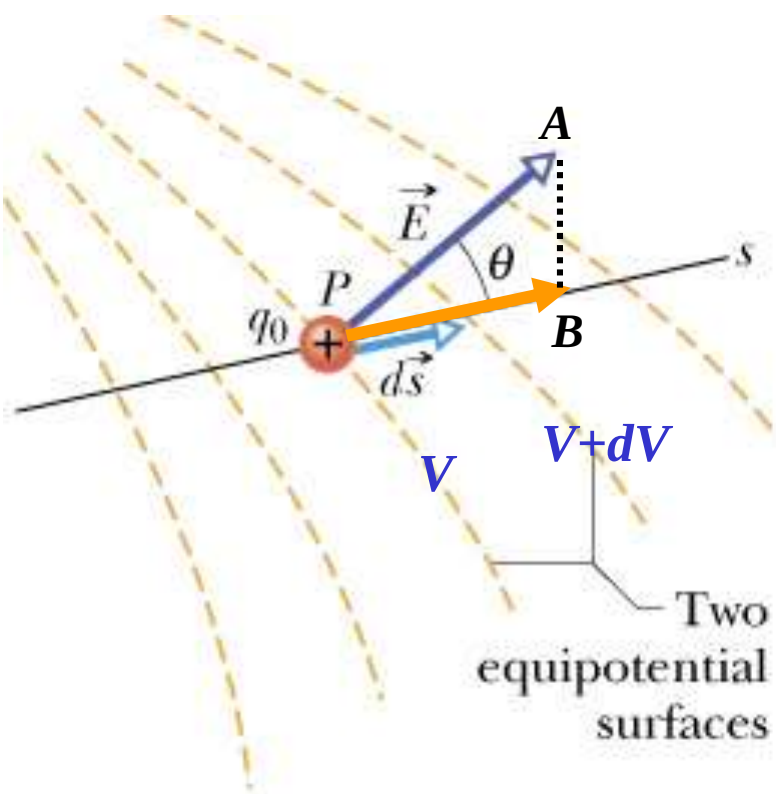
and radius $r = \dfrac{q}{4\pi\varepsilon_0 V}$.

(24-12)

Calculating the Electric Field $\vec{E}$ from the Potential V

Now we will tackle the reverse problem, i.e., determine $\vec{E}$ if we know V.

Consider two equipotential surfaces that correspond to the values V and $V + dV$ separated by a distance ds as shown in the figure. Consider an arbitrary direction represented by the vector $d\vec{s}$. We will allow the electric field to move a charge q_0 from the equipotential surface V to the surface $V + dV$.



The work done by the electric field is given by:

$W = -q_0 dV$ (eq. 1).

Also $W = Fds\cos\theta = Eq_0 ds\cos\theta$ (eq. 2)

If we compare these two equations we have:

$$Eq_0 ds\cos\theta = -q_0 dV \rightarrow E\cos\theta = -\frac{dV}{ds}.$$

From triangle PAB we see that $E\cos\theta$ is the component E_s of $\vec{E}$ along the direction s.

Thus: $E_s = -\dfrac{\partial V}{\partial s}$.

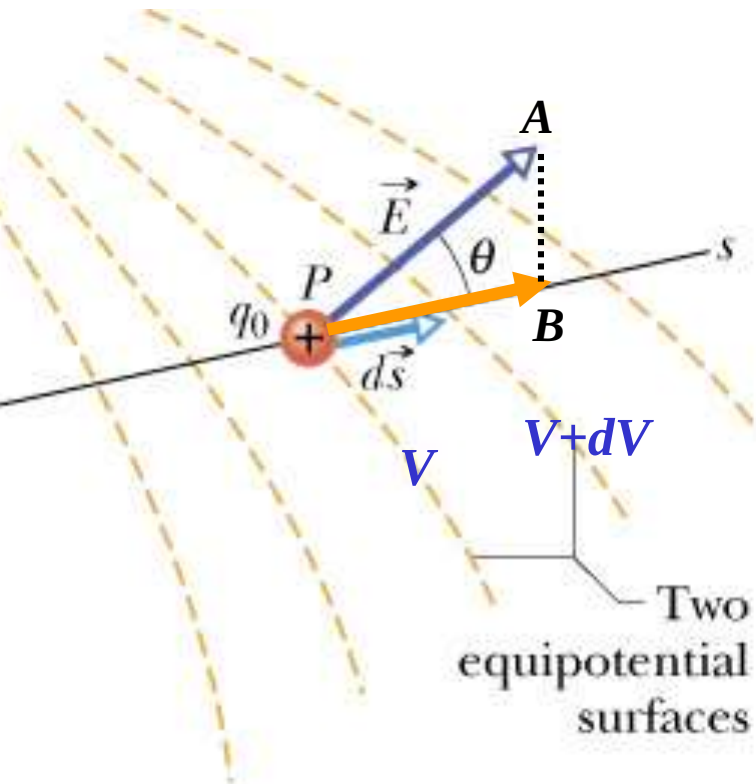
$$\boxed{E_s = -\frac{\partial V}{\partial s}}$$

(24-13)

$$E_s = -\frac{\partial V}{\partial s}$$

We have proved that $E_s = -\dfrac{\partial V}{\partial s}$.

The component of $\vec{E}$ in any direction is the negative of the rate at which the electric potential changes with distance in this direction.



If we take s to be the x- , y-, and z-axes we get:

$$E_x = -\frac{\partial V}{\partial x}$$

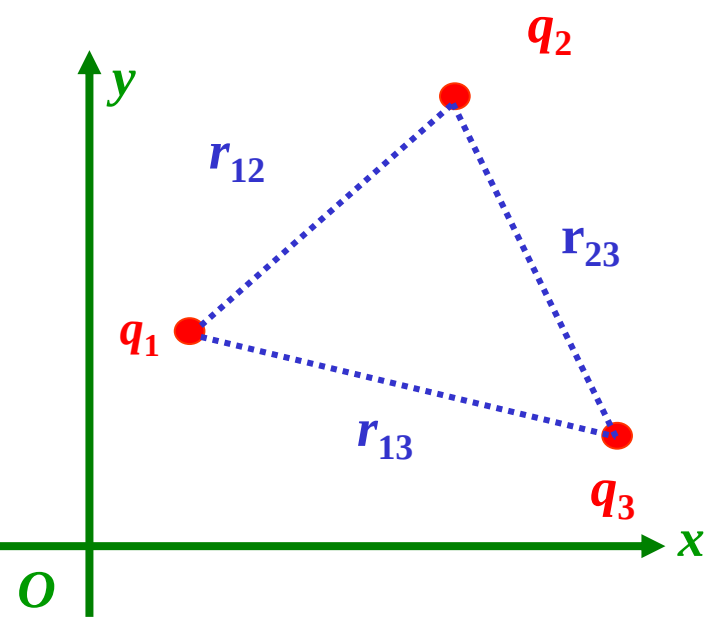
$$E_y = -\frac{\partial V}{\partial y}$$

$$E_z = -\frac{\partial V}{\partial z}$$

If we know the function $V(x, y, z)$ we can determine the components of $\vec{E}$ and thus the vector $\vec{E}$ itself :

$$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$$

(24-14)



Potential Energy U of a System of Point Charges

We define U as the work required to assemble the system of charges one by one, bringing each charge from infinity to its final position.

Using the above definition we will prove that for a system of three point charges U is given by:

$$\boxed{U = \frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}} + \frac{q_2 q_3}{4\pi\varepsilon_0 r_{23}} + \frac{q_1 q_3}{4\pi\varepsilon_0 r_{13}}}$$

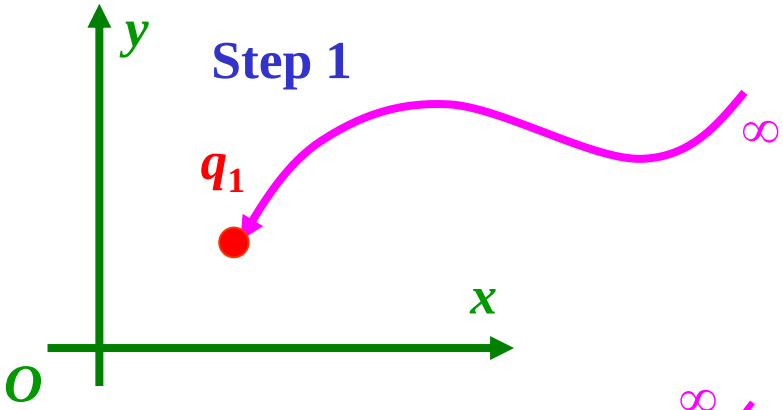
Note : Each pair of charges is counted only once.

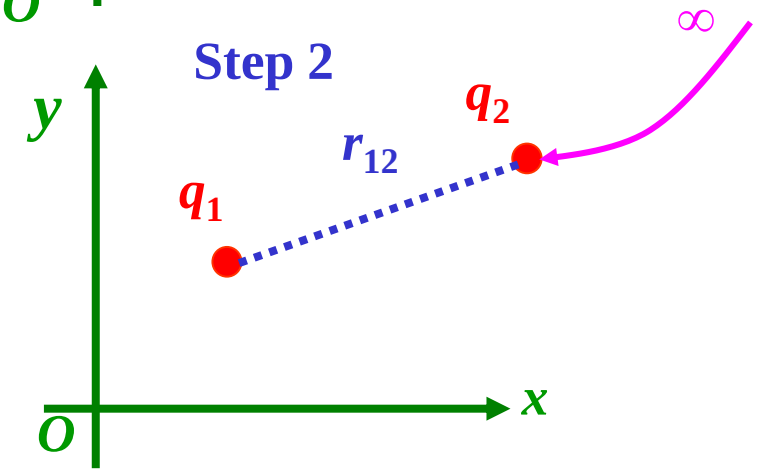
For a system of n point charges $\{q_i\}$ the potential energy U is given by:

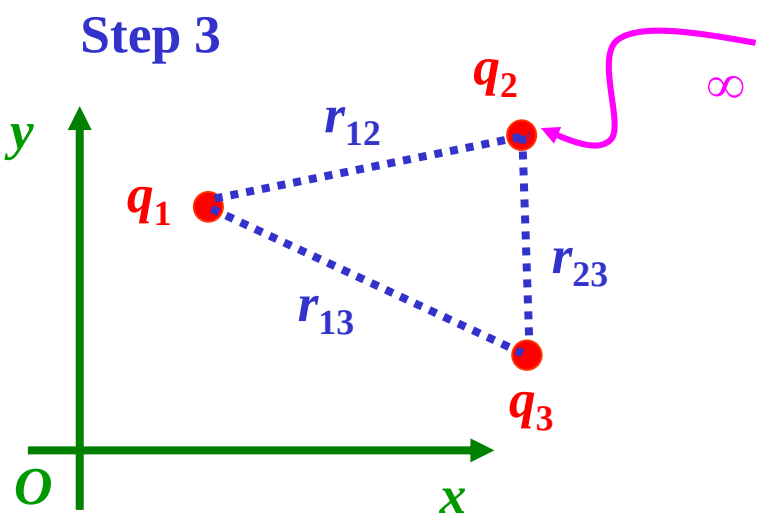
$$U = \frac{1}{4\pi\varepsilon_0} \sum_{\substack{i,j=1 \\ i<j}}^{n} \frac{q_i q_j}{r_{ij}} .$$

Here r_{ij} is the separation between q_i and q_j.

The summation condition $i < j$ is imposed so that, as in the case of three point charges, each pair of charges is counted only once.







Step 1 : Bring in q_1:

$$W_1 = 0$$

(no other charges around)

Step 2 : Bring in q_2:

$$W_2 = q_2 V(2)$$

$$V(2) = \frac{q_1}{4\pi\varepsilon_0 r_{12}} \rightarrow W_2 = \frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}}$$

Step 3 : Bring in q_3:

$$W_3 = q_3 V(3)$$

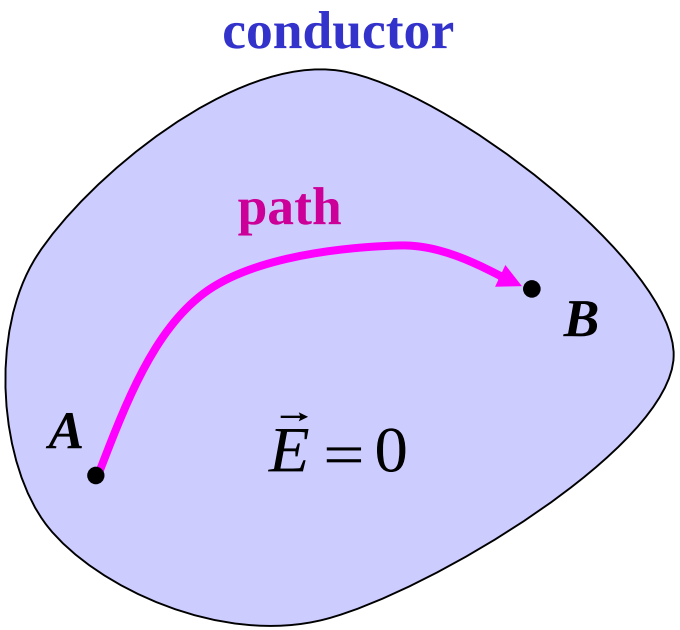
$$V(3) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}}\right) \rightarrow$$

$$W_3 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right)$$

$$W = W_1 + W_2 + W_3$$

$$W = \frac{q_1 q_2}{4\pi\varepsilon_0 r_{12}} + \frac{q_2 q_3}{4\pi\varepsilon_0 r_{23}} + \frac{q_1 q_3}{4\pi\varepsilon_0 r_{13}}$$

(24-16)



Potential of an Isolated Conductor

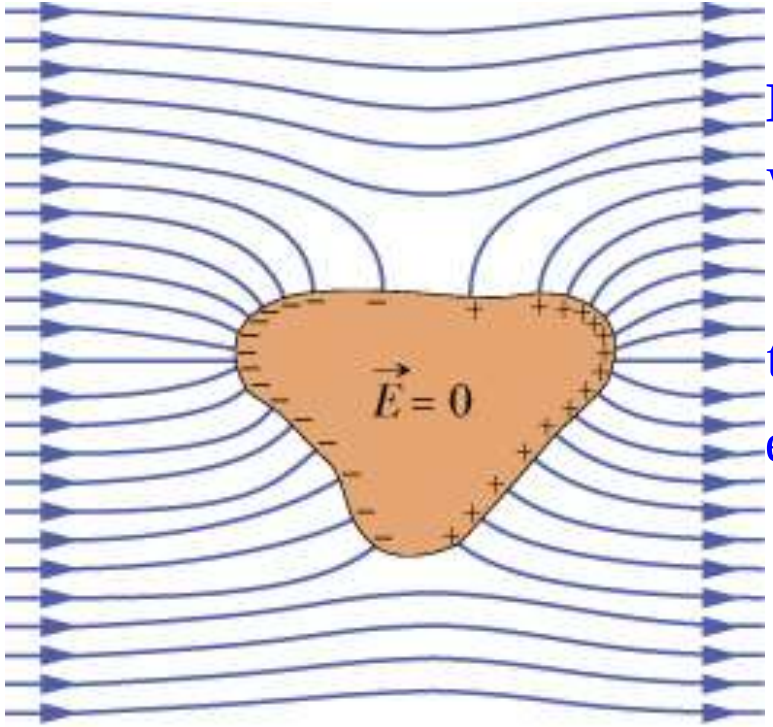
We shall prove that all the points on a conductor (either on the surface or inside) have the same potential.

A conductor is an equipotential surface.

Consider two points A and B on or inside a conductor. The potential difference $V_B - V_A$ between these two points is given by the equation:

$$V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{S}.$$

We already know that the electrostatic field $\vec{E}$ inside a conductor is zero. Thus the integral above vanishes, and $V_B = V_A$ for any two points on or inside the conductor.



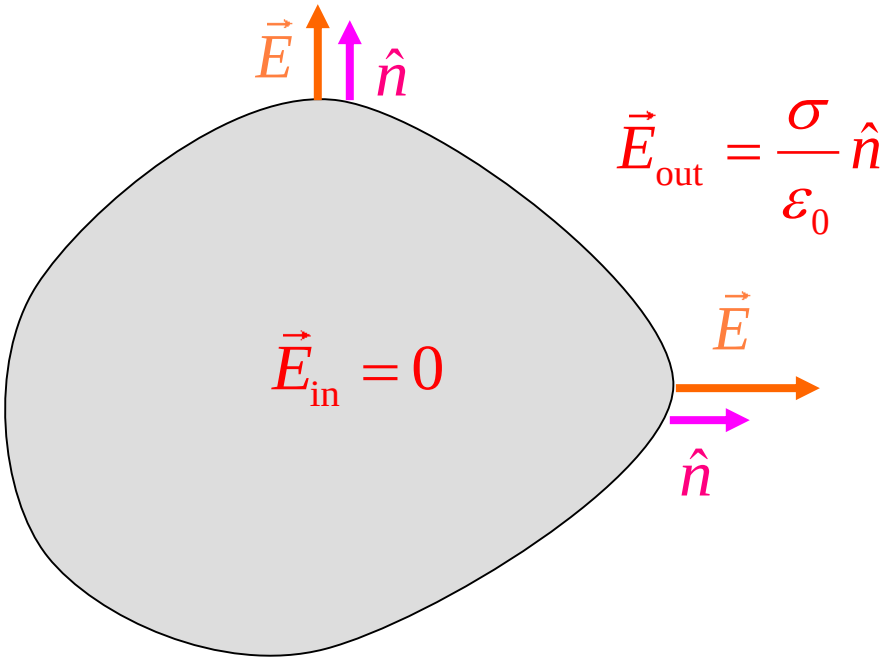
Isolated Conductor in an External Electric Field

We already know that the surface of a conductor is an equipotential surface. We also know that the electric field lines are perpendicular to the equipotential surfaces.

From these two statements it follows that the electric field vector $\vec{E}$ is perpendicular to the conductor surface, as shown in the figure.

All the charges of the conductor reside on the surface and arrange themselves in such a way so that the net electric field inside the conductor $E_{\text{in}} = 0$.

The electric field just outside the conductor is: $E_{\text{out}} = \dfrac{\sigma}{\varepsilon_0}$. **(24-18)**



Electric Field and Potential in and around a Charged Conductor : A Summary

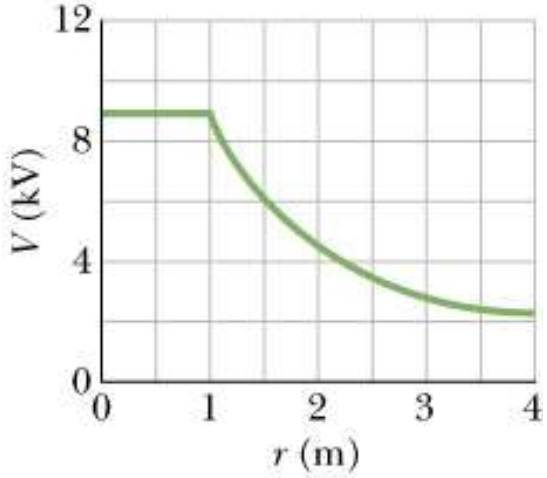
1. All the charges reside on the conductor surface.

2. The electric field inside the conductor is zero: $E_{\text{in}} = 0.$

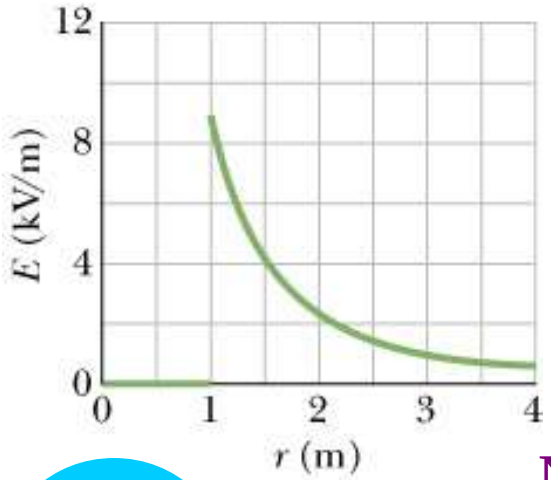
3. The electric field just outside the conductor is: $E_{\text{out}} = \dfrac{\sigma}{\varepsilon_0}.$

4. The electric field just outside the conductor is perpendicular to the conductor surface.

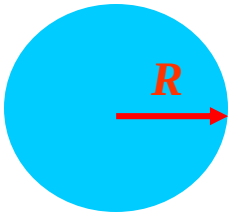
5. All the points on the surface and inside the conductor have the same potential. The conductor is an equipotential surface.



(a)



(b)



Electric Field and Electric Potential for a Spherical Conductor of Radius *R* and Charge *q*

For $r < R$, $V = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{R}$.

For $r > R$, $V = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r}$.

For $r < R$, $E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{R^2}$.

For $r > R$, $E = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r^2}$.

Note : Outside the spherical conductor, the electric field and the electric potential are identical to that of a point charge equal to the net conductor charge and placed at the center of the sphere.